

# iGauchoback Project

## Session Work Summary

Date: May 23, 2026

Production: <https://hackathon-nu-taupe.vercel.app>

Repository: acadmap/ (GitHub: Hackathon)

### 1. Executive Summary

This session completed the UCSB College of Engineering (CoE) Wave 2 pillar, upgraded interactive roadmap UX across engineering and L&S majors, generated interactive graph seeds for all 58 Letters & Science majors, and rebranded the product from AcadMap to iGauchoback. All changes were validated with `npm run build`, committed to main, and deployed to Vercel production.

### 2. CoE Wave 2 - GEAR Roadmaps & Quarter Plans

#### Infrastructure & data model

- Added `coe-major-detail-types.ts`, `loadCoeMajorDetail()`, `gear_framework` in `coe-catalog.json`
- Created `coe-majors/*.json` for all 5 CoE majors from GEAR 25-26 GRID tables
- Added `gear-sources.json`, `CoeQuarterTimeline`, `coe-planning.ts` (CoE GE injection)
- Scripts: `parse:gear`, `sync:coe-details`, `validate:coe-details`
- Docs: COE-GEAR-AGENT-CLUSTERS.md, updated AGENTS.md and GEAR-25-26.md

#### Interactive roadmaps (5 CoE majors)

- Chemical Engineering, Computer Engineering, Computer Science, Electrical Engineering, Mechanical Engineering
- All majors: `roadmap_available` true, GEAR detail pages, hub badges
- Commit: 6d5ec77 - Complete CoE Wave 2

### 3. Roadmap UX Improvements

#### Node spacing & overlap fix

- Re-laid out seed JSON with 300px column gaps and 220px row gaps
- Added `lib/roadmap-layout.ts`: `spreadRoadmapNodePositions()` + `digit-column` layout
- Runtime layout applied in `lib/flow.ts` so cards with descriptions never overlap

#### Click-to-focus interaction

- Selected node highlights; all other nodes dim (30% opacity, grayscale)
- Unrelated edges fade; connected edges stay visible; canvas background dims
- Click canvas or same node again to clear focus

#### Course descriptions

- Descriptions on node cards (truncated) and full text in sidebar
- Sourced from GEAR, UCSBPlat (CMPSC/ECE), and Is-majors detail files

### 4. Engineering Major Roadmap Updates

- Electrical Engineering - full relayout + UCSBPlat descriptions (4894020)
- Computer Science - spacing + descriptions from UCSBPlat CMPSC (fe5e902)
- ChE, CMPE, ME seeds exist; benefit from shared layout engine

### 5. L&S Interactive Roadmaps - All 58 Majors

### Phase 1: First 20 majors (deafed8)

- Actuarial Science (hand-curated) + 18 generated seeds
- Added scripts/generate-ls-seeds.ts and npm run generate:ls-seeds

### Phase 2: Majors 21-40 (de69383)

- Economics through Linguistics; FMS kept curated seed
- Hand-built History of Policy, Law & Governance (sparse catalog data)

### Phase 3: Majors 41-58 (3f0a39a)

- Mathematics through Zoology; Political Science via catalog fallback for POL S codes
- All 58 L&S catalog entries: roadmap\_available true, Live graph badges on hub

### Each L&S roadmap includes

- 8-18 course nodes from ls-majors detail + career outcome nodes
- Prerequisite chain edges + leads\_to career edges
- Non-overlapping layout and focus mode (shared graph components)

## 6. Rebrand: AcadMap to iGauchBack

- New lib/brand.ts centralizes APP\_NAME, APP\_TAGLINE, pageTitle(), theme key
- Navbar logo: AM to iG; all page titles and metadata updated
- Internal: iGauchBackNodeData, igauchoback-flow CSS, igauchoback-theme localStorage
- npm package renamed to igauchoback (d19c7de)
- Removed accidental .tmp\_pdfs from repo; added to .gitignore (47217cf)

## 7. Key File Paths

Data: data/seeds/ucsb-{slug}.json (63 roadmap seeds)

CoE details: data/ucsb/coe-majors/

L&S details: data/ucsb/ls-majors/ (58 files)

Graph UI: components/RoadmapGraph.tsx, RoadmapView.tsx, NodeCard.tsx, Sidebar.tsx

Layout: lib/roadmap-layout.ts, lib/flow.ts

Branding: lib/brand.ts

Generator: scripts/generate-ls-seeds.ts

## 8. npm Scripts Added / Used

- parse:gear, sync:coe-details, validate:coe-details
- generate:ls-seeds, relayout:seeds, relayout:seeds:ls
- sync:details (L&S catalog roadmap flags)

## 9. Deployment Commits (main branch)

- 6d5ec77 - CoE Wave 2 - GEAR quarter plans + 5 CoE roadmaps
- 4894020 - EE roadmap spacing, focus mode, descriptions
- fe5e902 - CS roadmap spacing + UCSBPlat descriptions
- 278f381 - L&S auto-layout + actuarial/FMS spacing
- deafed8 - L&S interactive roadmaps 1-20
- de69383 - L&S interactive roadmaps 21-40
- 3f0a39a - L&S interactive roadmaps 41-58 (complete)
- d19c7de - Rebrand to iGauchBack
- 47217cf - Remove temp PDF artifacts from repo

## 10. Success Criteria Met

- All 5 CoE majors: GEAR detail pages + quarter timelines + live roadmaps
- All 58 L&S majors: live interactive roadmaps at /roadmap/ucsb/{slug}
- npm run build passes; Vercel production deploys successful
- CoE hub: GEAR detail + Live graph badges on all 5 cards
- L&S hub: Live graph on all 58 cards

## 11. Suggested Next Steps

- Upgrade catalog-tier L&S seeds to sheet fidelity (department PDF QA)
- Hand-curate high-traffic majors (Econ, CS-adjacent L&S, Biology) with richer prereq graphs
- Faculty sidebar integration (Phase 3 docs) across more departments
- Transfer-student GRID variants for CoE (GEAR is freshman-focused)
- Custom domain / branding assets for iGachoBack (logo SVG, favicon)

## 12. Architecture Overview

Student flow: /schools/ucsb/letters-science or /schools/ucsb/engineering -> major card -> /roadmap/ucsb/{slug} (interactive React Flow graph). CoE majors also have GEAR detail pages with quarter-by-quarter timelines.

Data pipeline: ls-majors/\*.json + catalog -> generate-ls-seeds.ts -> data/seeds/. coe-majors/\*.json + GEAR PDFs -> parse-gear-grids.ts -> sync-coe-details.ts. seeds -> lib/flow.ts (layoutRoadmapNodes) -> RoadmapGraph.tsx.

## 13. Errors Resolved This Session

- FacultySidebarSection.tsx needed 'use client' directive (build fix)
- SeedRoadmapInput: added optional metadata field to lib/types.ts
- Political Science seed: fixed regex for multi-word dept codes (POL S 1)
- History of Policy/Law/Governance: hand-built seed from catalog sample electives
- Git: removed accidental .tmp\_pdfs staging; added to root .gitignore